Furqan Waris

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PROFESSIONAL SUMMARY

Full-Stack AI Engineer and Computer Science graduate skilled in bridging modern web architectures with applied artificial intelligence. Adept at building end-to-end, AI-powered applications from data ingestion and model integration to production-ready APIs.

EXPERIENCE

QTO Dev **Jan 2026 – Present**
AI Engineer *Lahore, Pakistan*

- Optimized PDF document ingestion using Docling and parallel execution, increasing document parsing throughput by 16% through concurrent processing.
- Developed an end-to-end RAG pipeline for construction estimation, retrieving relevant project specifications and requirements from construction documents; improved Precision@20 from 66% to 83% and increased Recall@20 by 10% through retrieval optimization.
- Programmed stateful workflow routing using LangGraph to handle complex user intents, implementing fallback loops and error-handling layers to gracefully manage database timeouts and off-topic queries.
- Integrated Langfuse for end-to-end LLM observability, tracking prompt latencies, costs, and token usage to optimize agent execution loops.

Techroniqs **Feb 2025 – Jan 2026**
Software Engineer *Lahore, Pakistan*

- Implemented asynchronous data fetching (AJAX / polling) for continuous server updates.
- Query Optimization: Implemented MongoDB indexing strategies to optimize query performance, reducing data retrieval times for large datasets and improving application responsiveness under high concurrent user load.

University of Central Punjab **May 2025 – Aug 2025**
Teaching Assistant *Lahore, Pakistan*

- Mentored 50+ students in programming fundamentals, logic-building, and flowchart design, improving lab performance by 20%.
- Assisted students in debugging C/C++ and Python programs, focusing on logic errors and performance optimization.
- Participated in code reviews, sprint discussions, and collaborative problem-solving environments.

AK Soft **Feb 2024 – Aug 2024**
Frontend Developer *Lahore, Pakistan*

- Built and maintained responsive web interfaces using React, Next.js, and Tailwind CSS, delivering consistent UI across multiple devices and screen sizes.
- Optimized page load performance through code-level improvements, resulting in faster load times and a smoother user experience.

PROJECTS

Galaxy Knowledge LLM Fine-Tuning *Python, LangGraph, Langfuse*

- Fine-tuned a Llama-based LLM on a domain-specific galaxy knowledge dataset using Python, PyTorch, Hugging Face Transformers, and LoRA/PEFT to improve responses to astronomy-related queries.
- Built the data preprocessing, fine-tuning, and evaluation pipeline, comparing the fine-tuned model against the base model on unseen galaxy-related question with Error rate 65% to 12 %.

Enterprise ERP AI Agent *Python, LangGraph, Langfuse*

- Architected an agentic ERP assistant using LangGraph, enabling autonomous multi-step reasoning, stateful workflows, and dynamic tool calling across enterprise systems.
- Integrated Langfuse for end-to-end LLM observability, prompt tracking, and latency optimization, boosting operational productivity by 40%.

AI-Powered HR Recruitment System *Python, MongoDB, RAG, Semantic Search*

- Developed a machine learning-driven screening application to match candidate resumes with job specifications using semantic search, hybrid retrieval, and custom text processing pipelines.
- Modeled structured talent profiles and job data using MongoDB & built optimized backend APIs to execute fast, responsive candidate searches.
- Enhanced candidate filtering accuracy and reduced initial CV screening times, showcasing strong software engineering principles and domain-specific problem-solving.

AI-Powered Intelligent Web Scraping Platform *Python, MongoDB*

- Developed an AI-driven web scraper using Playwright and FastAPI to dynamically crawl, parse, and ingest project listings and bidding opportunities across 1,000+ target websites.
- Resolved high selector failure rates across diverse, changing web layouts by building an automated error-analysis pipeline; monitored error rates and exception types to dynamically adjust scraping logic and self-correct parsing paths.

Voice Agent *FastAPI, Uplift AI, RAG (AI Agent Platform)*

- Built a dynamic voice-agent platform using Uplift AI, enabling users to create and configure personalized voice assistants through natural, two-way conversations.
- Implemented agent-specific guardrails, identity, and knowledge configuration with a FastAPI backend and RAG-based architecture, allowing each generated assistant to operate with customized instructions and domain knowledge.

AI-Powered E-Commerce RAG & LLMOps Platform *Weaviate, Agentic RAG, Phoenix*

- Architected and deployed an end-to-end Agentic RAG system featuring intelligent query routing, self-correction loops, and recursive retrieval using Python and Weaviate, boosting **precision** by 8% and **recall** by 14%.
- Deployed automated evaluation pipelines utilizing Phoenix AI Observability, OpenTelemetry, and Flask to monitor latency, trace complex agentic reasoning paths, and measure retrieval faithfulness across multi-turn product queries.

AI-Driven Automated Grading Platform *Node.js, MongoDB, Mongoose (Full-Stack EdTech Project)*

- Integrated LLM-based code evaluation constrained by deterministic scoring rules to ensure fair, consistent, and bias-reduced grading.
- Modeled grading tasks and metadata using MongoDB & Mongoose, ensuring traceable and auditable grading records.
- Integrated a Python-based AI grading engine with Node.js to process extracted code and apply course-specific rubrics.

AWS Data Ingestion and ETL Pipeline *AWS S3, Glue, EC2, ETL Pipelines*

- Designed and implemented batch data ingestion workflows using AWS services.
- Performed data extraction, transformation, and loading (ETL) operations on structured datasets.
- Applied data quality validation and monitoring techniques to ensure pipeline reliability.

Deep Learning System for Early-Stage Melanoma Detection *PyTorch, EfficientNet-B4 (Medical AI)*

- Developed and benchmarked multiple deep learning architectures trained from scratch (CNN custom variants, VGG, AlexNet, and ResNet) to classify skin lesions into clinical taxonomies.
- Achieved 87.49% classification accuracy, 86.09% recall, 80.32% F1-score, and 0.946 AUC using EfficientNet-B4.
- Built a data pipeline integrating ISIC Archive and HAM10000 datasets while addressing class imbalance.

Breast Cancer Diagnostic Predictive Model *Scikit-Learn, Random Forests, SVM*

- Integrated multiple supervised learning algorithms, including Random Forests, SVM, and XGBoost to classify cellular tumor characteristics as benign or malignant.
- Preprocessed tumor diagnostic data using Pandas and Scikit-Learn, implementing feature scaling, train-test splitting, and reproducible machine learning pipelines.
- Integrated statistical dimensionality reduction using Principal Component Analysis (PCA) to eliminate feature redundancy and optimize machine learning model accuracy.

Full-Stack Marketplace Infrastructure *MongoDB, Express.js, React, Node.js, JWT*

- Engineered a comprehensive property-sharing marketplace, enabling seamless end-to-end user journeys from discovery to checkout by building a MERN stack architecture with integrated booking management logic.
- Developed a secure, multi-role authentication system ensuring data isolation between hosts and guests by utilizing JWT-based session management and Express middleware.

EDUCATION

University of Central Punjab **Graduated with Honors**
Bachelor of Computer Science (CGPA: 3.49) *Lahore, Pakistan*

CERTIFICATIONS

- **Introduction to Data Engineering** | DeepLearning.AI & AWS
- **Machine Learning Specialization** | DeepLearning.AI & Stanford Online
- **Source Systems, Data Ingestion, and Pipelines** | DeepLearning.AI & AWS
- **AI Agent Developer Specialization** | Vanderbilt University
- **Data Collection and Processing with Python** | University of Michigan
- **Trustworthy Generative AI** | Vanderbilt University

LEADERSHIP & VOLUNTEERING

Member, IBTADA School **Nov 2025 – Feb 2026**
Youth Mentorship & Community Volunteering *Lahore, Pakistan*

- Volunteered in women empowerment and charity fundraising campaigns.
- Mentored school-level students in foundational problem-solving and academic skills.

TECHNICAL SKILLS

Programming Languages: JavaScript, Python, C, C++

Frontend: React.js, HTML5, CSS3

Backend: Node.js, Express.js

Databases: MongoDB, SQL

APIs: RESTful APIs

Authentication & Security: JWT, AES Encryption

Tools & Practices: Git, GitHub, Agile, UML Modeling, LangGraph, Playwright, n8n, Automation, S3, Glue, EC2, Power BI